

3(a)

```
clc, clearvars
```

```
fmd= readtable("FMD.xlsx", 'VariableNamingRule','preserve');  
east = fmd(fmd.EasternTurkey==1,:);  
west = fmd(fmd.EasternTurkey==0,:);  
fprintf("Summary of FMD1998 east and west respectively \n")
```

Summary of FMD1998 east and west respectively

```
summary(east.FMD1998)
```

26×1 double

NumMissing	0
Min	0
Median	0
Max	9
Mean	1.1923
Std	2.4003

```
summary(west.FMD1998)
```

40×1 double

NumMissing	0
Min	0
Median	0
Max	5
Mean	1.0750
Std	1.6546

```
fprintf("Summary of Cattle east and west respectively")
```

Summary of Cattle east and west respectively

```
summary(east.Cattle)
```

26×1 double

NumMissing	0
Min	43670
Median	121675
Max	622650
Mean	170080
Std	1.4810e+05

```
summary(west.Cattle)
```

40×1 double

NumMissing	0
Min	39330
Median	135025
Max	392840
Mean	157850
Std	8.2619e+04

3(b)

```
fmd.lnCattle= log(fmd.Cattle);  
mdl= fitglm(fmd, 'FMD1998 ~ lnCattle+ EasternTurkey',...
```

```
'Distribution','poisson','Link','log')
```

```
mdl =  
Generalized linear regression model:  
log(FMD1998) ~ 1 + EasternTurkey + lnCattle  
Distribution = Poisson
```

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-11.254	2.2801	-4.9359	7.9762e-07
EasternTurkey	0.041334	0.23936	0.17268	0.8629
lnCattle	0.94682	0.18828	5.0287	4.9379e-07

```
66 observations, 63 error degrees of freedom  
Dispersion: 1  
Chi^2-statistic vs. constant model: 26.3, p-value = 1.96e-06
```

```
ci= coefCI(mdl)
```

```
ci = 3x2  
-15.8105 -6.6979  
-0.4370 0.5197  
0.5706 1.3231
```

3(d)

```
exp_fmd= predict(mdl, table(log(170080),1,...  
    'VariableNames',{'lnCattle','EasternTurkey'}));  
fprintf('\nExpected FMD in Eastern (Cattle=170080): %.3f\n', exp_fmd);
```

Expected FMD in Eastern (Cattle=170080): 1.210

3(e)

```
p_le2_east = poisscdf(2, exp_fmd);  
fprintf('P(Y<=2) in Eastern: %.4f\n', p_le2_east);
```

P(Y<=2) in Eastern: 0.8773

3(f)

```
mu_west = predict(mdl, table(log(157850), 0, ...  
    'VariableNames',{'lnCattle','EasternTurkey'}));  
fprintf('Expected FMD in Western (Cattle=157850): %.3f\n', mu_west);
```

Expected FMD in Western (Cattle=157850): 1.082

3(g)

```
p_le2_west = poisscdf(2, mu_west);  
fprintf('P(Y<=2) in Western: %.4f\n', p_le2_west);
```

P(Y<=2) in Western: 0.9040